

scPTR: Decomposing Post-Transcriptional Regulation at Single-Cell Resolution

Introduction

Transcript abundance is jointly determined by synthesis and degradation, yet single-cell computational frameworks have addressed only one side of this balance. RNA velocity models transcriptional dynamics from spliced/unspliced ratios [1,2], but mRNA degradation—orchestrated by RNA-binding proteins (RBPs), microRNAs, and *cis*-regulatory elements—is an equally fundamental determinant of steady-state expression and a recognized driver of cell fate transitions [3]. No framework exists to estimate per-cell, per-gene degradation rates, discover populations defined by post-transcriptional programs, or infer the regulatory networks controlling mRNA stability at single-cell resolution. We present **scPTR** (Single-Cell Post-Transcriptional Regulatory Decomposition), a framework that makes per-cell mRNA degradation rates (γ) a first-class observable from standard scRNA-seq—complementary to, and independent of, RNA velocity.

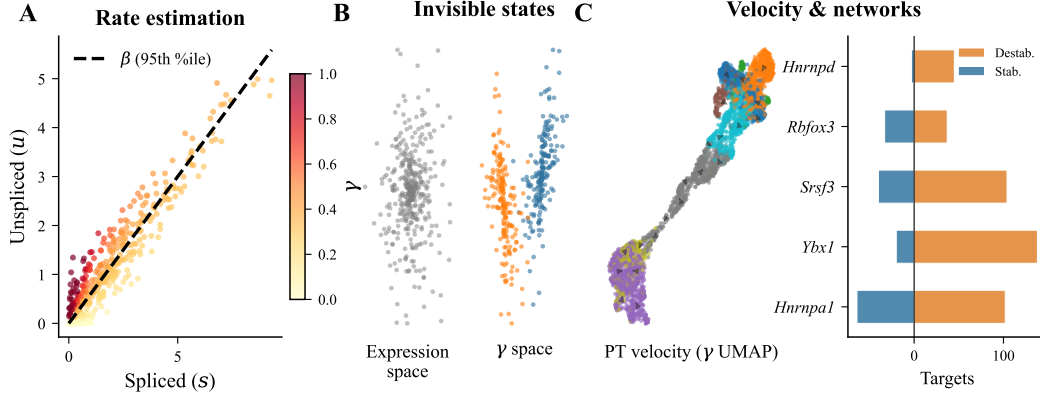


Figure 1: scPTR framework. (A) Gene-specific splicing rates (β) are estimated via quantile regression on unspliced/spliced phase portraits; per-cell degradation rates are then computed as $\gamma = \beta \cdot u/s$ after k -NN Gaussian-kernel smoothing. (B) Clustering in γ -space via PCA and Leiden community detection reveals post-transcriptional states that are invisible to standard expression-based analysis. (C) Left: post-transcriptional velocity field on the γ -space UMAP of pancreas cells (3,696 cells, 8 cell types). Right: top RBP hubs from library-size-corrected network inference, showing destabilizing and stabilizing target counts.

Methods

For each gene, scPTR estimates the splicing rate β as the 95th-percentile slope of the unspliced/spliced phase portrait (Fig. 1A), then computes per-cell degradation rates via the kinetic steady-state relation $\gamma_g = \beta_g \cdot u_{ig}/s_{ig}$, where u and s are k -NN Gaussian-kernel-smoothed unspliced and spliced counts, respectively. The smoothing step uses adaptive bandwidths (median neighbor distance per cell) to denoise inherently sparse unspliced counts without oversmoothing. Post-transcriptional (PT) states are discovered by performing PCA, k -NN graph construction, and Leiden clustering entirely in γ -space rather than expression space (Fig. 1B). PT velocity is defined as the weighted mean γ difference between each cell and its neighbors, yielding a per-cell vector field orthogonal to RNA velocity (Fig. 1C). RBP–target regulatory networks are inferred via elastic net regression of each target gene’s γ on smoothed RBP expression, with library-size partial correlation to correct for a systematic destabilizing bias arising from the correlation between library size and count-based γ estimates. We additionally introduce DeepPTR, a structured variational autoencoder with a kinetic decoder that factorizes the latent space into transcriptional (z_T) and post-transcriptional (z_{PT}) components under a negative binomial observation model.

Results

We validated γ estimates against published mRNA half-lives across three species–platform combinations. On 10x developmental data, scPTR (Spearman $\rho = -0.40$, pancreas) outperforms scVelo steady-state ($\rho = -0.37$) and velVI ($\rho = -0.28$); in the sci-fate metabolic-labeling dataset, where 4sU directly quantifies new/old RNA, ρ reaches -0.81 (Fig. 2A). Consistent with known *cis*-regulatory determinants, γ correlated positively with 3’ UTR length ($\rho = 0.34$, $p < 10^{-200}$) and AU content ($\rho = 0.30$, $p < 10^{-150}$), and miRNA targets from TargetScan 8.0 were enriched among high- γ genes for 59% of 215 tested families (Fig. 2B; aggregate $p = 4.7 \times 10^{-65}$). Estimates are robust to cell subsampling ($r > 0.97$ at 20%) and scale sub-linearly (~ 91 seconds estimated for 100,000 cells).

Clustering in γ -space revealed expression-invisible PT states in 3 of 8 pancreatic and 6 of 11 hippocampal cell types (Fig. 3A)—subpopulations with distinct degradation programs that are undetectable by expression analysis and confirmed non-artifactual by zero-permutation control (ARI ≈ 0). The Epsilon cluster showed the strongest invisibility: silhouette = 0.215 in γ -space versus -0.011 in expression space. These invisible states were enriched for tissue-appropriate pathways: ER stress and autophagy in pancreatic endocrine cells; synaptic plasticity and spliceosome in hippocampal neurons.

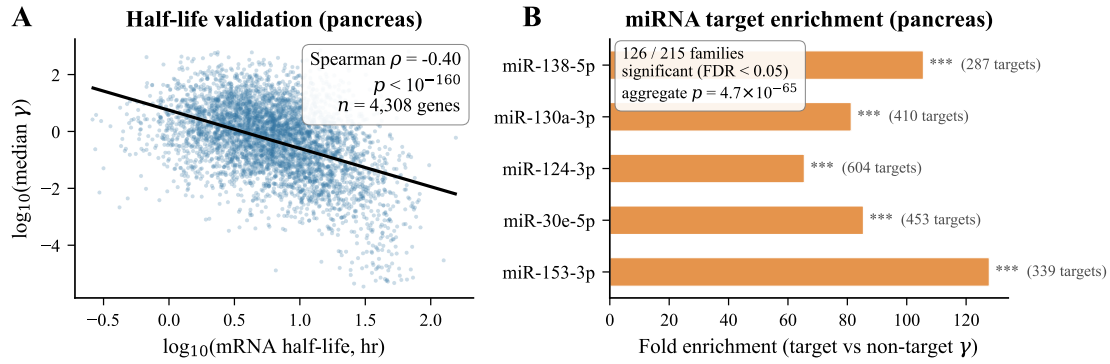


Figure 2: Validation of γ estimates. (A) Per-gene median γ correlates negatively with published mRNA half-lives in the pancreas dataset (Spearman $\rho = -0.40$, $n = 4,308$ genes; $p < 10^{-160}$ in sci-fate metabolic-labeling data), outperforming scVelo steady-state ($\rho = -0.37$) and velVI ($\rho = -0.28$). (B) Top miRNA families show strong fold enrichment of γ among predicted targets vs. non-targets in pancreas (126/215 families significant at FDR < 0.05 ; aggregate $p = 4.7 \times 10^{-65}$).

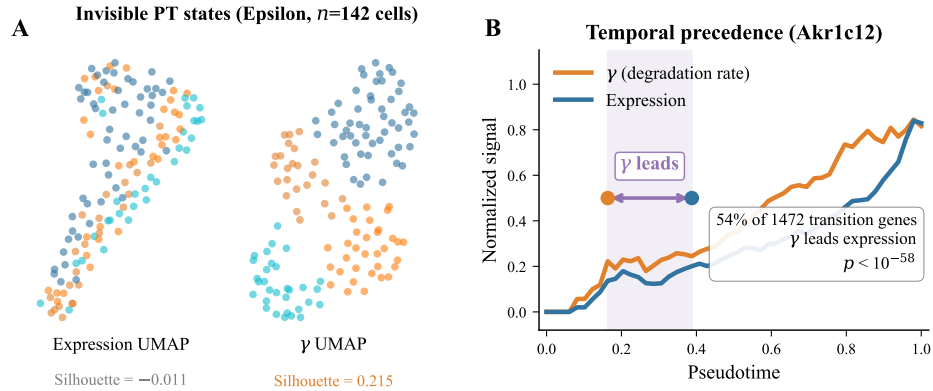


Figure 3: Key biological findings. (A) Pancreas Epsilon cells ($n = 142$) appear as one homogeneous cluster in expression-space UMAP (silhouette = -0.011) but separate into distinct post-transcriptional states in γ -space UMAP (silhouette = 0.215), demonstrating expression-invisible heterogeneity in mRNA degradation programs. (B) Along developmental pseudotime, changes in γ (red) precede changes in expression (blue) for 54% of 1,472 transition genes ($p < 10^{-57}$), shown here for *Akr1c12*. Post-transcriptional rewiring acts as an early fate-commitment signal.

PT velocity—the neighbor-averaged γ gradient—is orthogonal to RNA velocity (cosine similarity ≈ 0) and reveals that degradation-rate changes temporally precede expression changes: 54% of 1,472 transition genes in pancreas ($p < 10^{-57}$) and 78% of 146 genes in dentate gyrus ($p = 9.9 \times 10^{-13}$; Fig. 3B), implicating post-transcriptional rewiring as an early fate-commitment signal. Library-size-corrected RBP–target network inference identifies essential hub regulators (DepMap CRISPR; $p = 6.4 \times 10^{-5}$) including HNRNPA1, YBX1, and ELAVL1/HuR. Applied to human neuroblastoma [4], scPTR reveals a predominantly stabilizing RBP landscape (66% stabilizing edges after correction)—contrasting sharply with the destabilizing bias in developmental tissues and consistent with oncogenic mRNA stabilization. Hub RBPs (HNRNPA2B1, PABPC1) match known cancer dependencies [5].

Conclusions

scPTR establishes mRNA degradation rates as a measurable, biologically informative axis of single-cell analysis. It outperforms existing methods at half-life prediction, uncovers expression-invisible post-transcriptional cell states, demonstrates that degradation-rate rewiring precedes transcriptional changes during fate commitment, and identifies functionally essential RBP regulatory networks—opening a previously inaccessible layer of post-transcriptional biology to systematic single-cell investigation.

Availability: <https://github.com/bryanc5864/scPTR>

References

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